

Task Lecture 1 - Linux

1-Open your terminal and check your current working directory

```
mfadl237@DESKTOP-7BUEFV1 ~> pwd
/home/mfadl237
mfadl237@DESKTOP-7BUEFV1 ~> _
```

2. Create a directory structure under /tmp as follows:

/tmp/path_practice/dirA/dirB

```
mfadl237@DESKTOP-7BUEFV1 ~> sudo mkdir -p /tmp/path_practice/dirA/dirB
[sudo] password for mfadl237:
mfadl237@DESKTOP-7BUEFV1 ~>
```

3. From inside dirB , use:

A relative path to move back to dirA

An absolute path to move to /tmp/path_practice

```
mfadl237@DESKTOP-7BUEFV1 ~> cd /tmp/path_practice/
mfadl237@DESKTOP-7BUEFV1 /t/path_practice> pwd
/tmp/path_practice
mfadl237@DESKTOP-7BUEFV1 /t/path_practice> cd dirA/dirB/
mfadl237@DESKTOP-7BUEFV1 /t/p/d/dirB> cd ../
mfadl237@DESKTOP-7BUEFV1 /t/p/d/dirB> pwd
/tmp/path_practice/dirA/dirB
mfadl237@DESKTOP-7BUEFV1 /t/p/d/dirB> cd ../
mfadl237@DESKTOP-7BUEFV1 /t/p/dirA> pwd
/tmp/path_practice/dirA
mfadl237@DESKTOP-7BUEFV1 /t/p/dirA>
```

Question 2 — File and Directory Operations

1. Inside /tmp/linux_practice , create three empty files named:

file1.txt file2.txt file3.txt

```
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ ll
total 12
drwxr-xr-x  3 root root 4096 May 30 15:23 ./
drwxrwxrwt 10 root root 4096 May 30 15:26 ../
drwxr-xr-x  3 root root 4096 May 30 15:17 dirA/
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ ls -lh
total 4.0K
drwxr-xr-x 3 root root 4.0K May 30 15:17 dirA
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ pwd
/tmp/path_practice
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ sudo touch file{1..3}.txt
[sudo] password for mfadl237:
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ ls -lh
total 4.0K
drwxr-xr-x 3 root root 4.0K May 30 15:17 dirA
-rw-r--r-- 1 root root   0 May 30 15:27 file1.txt
-rw-r--r-- 1 root root   0 May 30 15:27 file2.txt
-rw-r--r-- 1 root root   0 May 30 15:27 file3.txt
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$
```

2. Create a new directory called backup

3. Copy all .txt files to the backup directory

```
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ ll
total 12
drwxr-xr-x  3 root root 4096 May 30 15:30 ./
drwxrwxrwt 10 root root 4096 May 30 15:26 ../
drwxr-xr-x  3 root root 4096 May 30 15:17 dirA/
-rw-r--r--  1 root root   0 May 30 15:27 file1.txt
-rw-r--r--  1 root root   0 May 30 15:27 file2.txt
-rw-r--r--  1 root root   0 May 30 15:27 file3.txt
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ sudo mkdir backup
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ cp file* backup/
cp: cannot create regular file 'backup/file1.txt': Permission denied
cp: cannot create regular file 'backup/file2.txt': Permission denied
cp: cannot create regular file 'backup/file3.txt': Permission denied
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ sudo cp file* backup/
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ cd backup/
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice/backup$ ll
total 8
drwxr-xr-x  2 root root 4096 May 30 15:32 ./
drwxr-xr-x  4 root root 4096 May 30 15:32 ../
-rw-r--r--  1 root root   0 May 30 15:32 file1.txt
-rw-r--r--  1 root root   0 May 30 15:32 file2.txt
-rw-r--r--  1 root root   0 May 30 15:32 file3.txt
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice/backup$
```

4. Rename file1.txt to notes.txt inside the main folder.

```
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice/backup$ cd ../
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ pwd
/tmp/path_practice
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ sudo mv file1.txt notes.txt
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ ll
total 16
drwxr-xr-x  4 root root 4096 May 30 15:34 ./
drwxrwxrwt 10 root root 4096 May 30 15:26 ../
drwxr-xr-x  2 root root 4096 May 30 15:32 backup/
drwxr-xr-x  3 root root 4096 May 30 15:17 dirA/
-rw-r--r--  1 root root   0 May 30 15:27 file2.txt
-rw-r--r--  1 root root   0 May 30 15:27 file3.txt
-rw-r--r--  1 root root   0 May 30 15:27 notes.txt
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$
```

5. Delete the file3.txt file only.

```
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ sudo rm file3.txt
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ ll
total 16
drwxr-xr-x  4 root root 4096 May 30 15:35 ./
drwxrwxrwt 10 root root 4096 May 30 15:26 ../
drwxr-xr-x  2 root root 4096 May 30 15:32 backup/
drwxr-xr-x  3 root root 4096 May 30 15:17 dirA/
-rw-r--r--  1 root root   0 May 30 15:27 file2.txt
-rw-r--r--  1 root root   0 May 30 15:27 notes.txt
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$
```

Question 3 — Practicing Vim or Nano

1. Open notes.txt in Vim.

```
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ sudo vim notes.txt
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$
```

2. Add three lines of text.

```

One Line
Two Line
Three Line
~
~
~
~
~
~

```

3. Use Vim commands to:

Copy the first line

.

yy

Paste it before the third line.

p

Delete the second line.

2d delete the current and next (2 Lines) – dd delete the current line

Save and exit.

:wq!

Question 4 — Text Viewing and Filtering

1. Use cat , head , and tail on /etc/passwd .

Display only the first 3 lines and the last 2 lines.

```

mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ head -n 3 /etc/passwd
root:x:0:0:root:/root:/usr/bin/fish
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ tail -n 2 /etc/passwd
polkitd:x:990:990:User for polkitd:/:usr/sbin/nologin
mfadl237:x:1000:1000::/home/mfadl237:/usr/bin/bash
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$

```

```
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ cat /etc/passwd
root:x:0:0:root:/root:/usr/bin/fish
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534:./nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:./usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:./usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101:./nonexistent:/usr/sbin/nologin
syslog:x:102:102:./nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:./usr/sbin/nologin
uidd:x:103:103:./run/uidd:/usr/sbin/nologin
landscape:x:104:105:./var/lib/landscape:/usr/sbin/nologin
polkitd:x:990:990:User for polkitd:./usr/sbin/nologin
mfad1237:x:1000:1000:./home/mfad1237:/usr/bin/bash
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$
```

2. Find all lines in /etc/passwd containing bash .

```
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ grep "bash" /etc/passwd
mfad1237:x:1000:1000:./home/mfad1237:/usr/bin/bash
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ grep -c "bash" /etc/passwd
1
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ grep -n "bash" /etc/passwd
28:mfad1237:x:1000:1000:./home/mfad1237:/usr/bin/bash
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$
```

3. Display all lines not containing nologin

```
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ grep -v "nologin" /etc/passwd
root:x:0:0:root:/root:/usr/bin/fish
sync:x:4:65534:sync:/bin:/bin/sync
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
mfad1237:x:1000:1000:./home/mfad1237:/usr/bin/bash
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ grep -vn "nologin" /etc/passwd
1:root:x:0:0:root:/root:/usr/bin/fish
5:sync:x:4:65534:sync:/bin:/bin/sync
21:dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
28:mfad1237:x:1000:1000:./home/mfad1237:/usr/bin/bash
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$
```

Question 5 — Exploring System Information and Saving Output

1. Display your current working directory and logged-in username.

```
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ pwd
/tmp/path_practice
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ whoami
mfadl237
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$
```

2. List all files and directories in your home directory using `ls -l -a` .

```
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ ls -la /home
total 12
drwxr-xr-x  3 root    root    4096 May 16 19:50 .
drwxr-xr-x 22 root    root    4096 May 29 15:13 ..
drwxr-x--- 13 mfadl237 mfadl237 4096 May 30 15:47 mfadl237
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$
```

3. Combine both commands and redirect the output into a file named `system_info.txt` inside `/tmp/linux_practice` .

```
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ pwd > system_info.txt
-bash: system_info.txt: Permission denied
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ pwd | sudo tee system_info.txt
/tmp/path_practice
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ cat system_info.txt
/tmp/path_practice
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$

mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ cat system_info.txt
/tmp/path_practice
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ whoami | sudo tee system_info.txt
mfadl237
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ cat system_info.txt
mfadl237
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ (pwd ; whoami) | sudo tee system_info.txt
/tmp/path_practice
mfadl237
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ cat system_info.txt
/tmp/path_practice
mfadl237
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$
```

4. View the saved file content using `less` and verify that both outputs are recorded

```
/tmp/path_practice
mfadl237
system_info.txt (END)
```

Question 6 — Using Pipes

1. Show all files in `/bin` that contain `sh` in their name.

2. Display only the first 5 results.

```
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ ls /bin | grep sh | head -n 5
bash
bashbug
byobu-shell
c_rehash
chsh
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$
```

Question 7 — Output Redirection

1. Redirect the output of `ls /etc` into `etc_list.txt` .

```
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ ls /etc | sudo tee etc_list.txt
PackageKit
X11
adduser.conf
alternatives
```

2. Append the current date to the same file.

```
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ date | sudo tee -a etc_list.txt
Sat May 30 16:12:44 EEST 2026
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ cat etc_list.txt
PackageKit
X11
adduser.conf
alternatives
```

3. Verify both outputs are saved using cat

```
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ tail -5 etc_list.txt
xattr.conf
xdg
xml
zsh_command_not_found
Sat May 30 16:12:44 EEST 2026
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$
```

Question 8 — Links

1. Create a file named main.txt and write "Linux is awesome!" inside it.

```
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ sudo touch main.txt
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ sudo vim main.txt
```

2. Create:

A hard link → main_hard.txt

A soft link → main_soft.txt


```
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ sudo ln main.txt main_hard.txt
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ ll
total 36
drwxr-xr-x 4 root root 4096 May 30 16:16 ./
drwxrwxrwt 9 root root 4096 May 30 16:06 ../
drwxr-xr-x 2 root root 4096 May 30 15:32 backup/
drwxr-xr-x 3 root root 4096 May 30 15:17 dirA/
-rw-r--r-- 1 root root 1673 May 30 16:12 etc_list.txt
-rw-r--r-- 1 root root 0 May 30 15:27 file2.txt
-rw-r--r-- 2 root root 17 May 30 16:14 main.txt
-rw-r--r-- 2 root root 17 May 30 16:14 main_hard.txt
-rw-r--r-- 1 root root 290 May 30 15:52 notes.txt
-rw-r--r-- 1 root root 28 May 30 16:05 system_info.txt
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ sudo ln -s main.txt main_soft.txt
[sudo] password for mfadl237:
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ ll
total 36
drwxr-xr-x 4 root root 4096 May 30 17:24 ./
drwxrwxrwt 9 root root 4096 May 30 16:35 ../
drwxr-xr-x 2 root root 4096 May 30 15:32 backup/
drwxr-xr-x 3 root root 4096 May 30 15:17 dirA/
-rw-r--r-- 1 root root 1673 May 30 16:12 etc_list.txt
-rw-r--r-- 1 root root 0 May 30 15:27 file2.txt
-rw-r--r-- 2 root root 17 May 30 16:14 main.txt
-rw-r--r-- 2 root root 17 May 30 16:14 main_hard.txt
lrwxrwxrwx 1 root root 8 May 30 17:24 main_soft.txt -> main.txt
-rw-r--r-- 1 root root 290 May 30 15:52 notes.txt
-rw-r--r-- 1 root root 28 May 30 16:05 system_info.txt
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$
```

3. Delete main.txt and check which link still works

```
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ sudo rm main.txt
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ ll
total 32
drwxr-xr-x 4 root root 4096 May 30 17:26 ./
drwxrwxrwt 9 root root 4096 May 30 16:35 ../
drwxr-xr-x 2 root root 4096 May 30 15:32 backup/
drwxr-xr-x 3 root root 4096 May 30 15:17 dirA/
-rw-r--r-- 1 root root 1673 May 30 16:12 etc_list.txt
-rw-r--r-- 1 root root 0 May 30 15:27 file2.txt
-rw-r--r-- 1 root root 17 May 30 16:14 main_hard.txt
lrwxrwxrwx 1 root root 8 May 30 17:24 main_soft.txt -> main.txt
-rw-r--r-- 1 root root 290 May 30 15:52 notes.txt
-rw-r--r-- 1 root root 28 May 30 16:05 system_info.txt
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ cat main_soft.txt
cat: main_soft.txt: No such file or directory
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$ cat main_hard.txt
Linux is awesome
mfadl237@DESKTOP-7BUEFV1:/tmp/path_practice$
```

Question 9 — Getting Help

1. Use man or --help to find what the rm -rf option does.

man rm

--help display this help and exit

-r, -R, --recursive

```

-r, -R, --recursive
    remove directories and their contents recursively

-d, --dir
    remove empty directories

-v, --verbose
    explain what is being done

--help display this help and exit

```

-f, --force
ignore nonexistent files and arguments, never prompt

```

-f, --force
    ignore nonexistent files and arguments, never prompt

```

remove directories and their contents recursively

2. Use man -k copy or tldr cp to check examples for copying files

```

mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ man -k cp
proc_cpuinfo (5)      - CPU and system architecture information
proc_pid_cpuset (5)   - CPU affinity sets
chcpu (8)             - configure CPUs
cp (1)               - copy files and directories
cp1251 (7)           - CP 1251 character set encoded in octal, decimal, and hexadecimal
cp1252 (7)           - CP 1252 character set encoded in octal, decimal, and hexadecimal
cpan (1)             - easily interact with CPAN from the command line
cpan5.38-x86_64-linux-gnu (1) - easily interact with CPAN from the command line
cpgr (8)             - copy with locking the given file to the password or group file
cppw (8)             - copy with locking the given file to the password or group file
cpuid (4)            - x86 CPUID access device
cpuset (7)          - confine processes to processor and memory node subsets
dhcpcd (8)           - a DHCP client
dhcpcd-run-hooks (8) - DHCP client configuration script
dhcpcd.conf (5)      - dhcpcd configuration file
docker-container-cp (1) - Copy files/folders between a container and the local filesystem
docker-cp (1)       - Copy files/folders between a container and the local filesystem
ecparam (1ssl)      - OpenSSL application commands
filesystems (5)     - Linux filesystem types: ext, ext2, ext3, ext4, hpfs, iso9660, JFS, minix, msdos,
fs (5)              - Linux filesystem types: ext, ext2, ext3, ext4, hpfs, iso9660, JFS, minix, msdos,
ip-mptcp (8)        - MPTCP path manager configuration
ip-tcp_metrics (8)  - management for TCP Metrics
lscpu (1)           - display information about the CPU architecture
msr (4)            - x86 CPU MSR access device
ss (1)             - arbitrary TCP and UDP connections and listeners

```

Question 10 — Challenge Task ☐

Find all log entries in /var/log/syslog that contain error (case-insensitive),
show only the last 5 lines, and save the output to /tmp/linux_practice/errors.txt .

```

mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ sudo grep -in "error" /var/log/syslog | tail -5 | sudo tee -a errors.txt
6:2026-05-30T15:05:25.660694+03:00 DESKTOP-7BUEFV1 kernel: WSL (191) ERROR: CheckConnection: getaddrinfo() failed: -3
7:2026-05-30T15:05:33.442079+03:00 DESKTOP-7BUEFV1 kernel: WSL (191) ERROR: CheckConnection: getaddrinfo() failed: -5
60:2026-05-30T15:18:36.182859+03:00 DESKTOP-7BUEFV1 kernel: WSL (191) ERROR: CheckConnection: getaddrinfo() failed: -5
227:2026-05-30T17:24:44.667858+03:00 DESKTOP-7BUEFV1 kernel: WSL (191) ERROR: CheckConnection: getaddrinfo() failed: -3
231:2026-05-30T17:30:48.698462+03:00 DESKTOP-7BUEFV1 kernel: WSL (191) ERROR: CheckConnection: getaddrinfo() failed: -3
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$ cat errors.txt
6:2026-05-30T15:05:25.660694+03:00 DESKTOP-7BUEFV1 kernel: WSL (191) ERROR: CheckConnection: getaddrinfo() failed: -3
7:2026-05-30T15:05:33.442079+03:00 DESKTOP-7BUEFV1 kernel: WSL (191) ERROR: CheckConnection: getaddrinfo() failed: -5
60:2026-05-30T15:18:36.182859+03:00 DESKTOP-7BUEFV1 kernel: WSL (191) ERROR: CheckConnection: getaddrinfo() failed: -5
227:2026-05-30T17:24:44.667858+03:00 DESKTOP-7BUEFV1 kernel: WSL (191) ERROR: CheckConnection: getaddrinfo() failed: -3
231:2026-05-30T17:30:48.698462+03:00 DESKTOP-7BUEFV1 kernel: WSL (191) ERROR: CheckConnection: getaddrinfo() failed: -3
mfad1237@DESKTOP-7BUEFV1:/tmp/path_practice$

```


